

CSMFAB000000000102 — Drone Redesign: Archimedes Screw Propulsion

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMFAB000000000102 – Drone Redesign: Archimedes Screw Propulsion V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Frame material	Carbon fiber (conductive, $\sigma \sim 10^3$ S/m)	BFRP/Elium® (insulating, $\rho > 10^{10}$ $\Omega \cdot m$)	Carbon fiber creates GIC loop; BFRP eliminates it at lower cost per kg	Engineering test data 2025; Elium® TDS
Rotor hub	Aluminum 6061-T6 (conductive)	Flash-sintered ZrB ₂ -SiC ceramic hub (1800°C, 5 min)	Al hub couples motor winding GIC to frame; ceramic hub breaks path	Johnson et al., Acta Materialia 278 (2024) 120223
EMI shielding on electronics bay	Copper mesh (SE ~45 dB)	Ti ₃ C ₂ T _x MXene spray 50 μm (SE 92 dB)	47 dB deficit in V1.0; MXene 2× protection, spray-applied vs mechanical mesh	ACS Appl. Mater. & Interfaces 17/8 (2025) 12234
Navigation	GPS-dependent	LoRa mesh 915 MHz 64-node + inertial dead-reckoning	GPS degraded by Solar Energetic Particles during Carrington event	ITU-R WP 4C 2024; LoRa Alliance v1.1

Parameter	V1.0	V2.0	Reason	Primary Source
Energy harvesting	None beyond main battery	PVDF-TrFE bimorph (lead-free) on rotor arms: 48 mW harvest	Lead-free mandate (RoHS); harvesting extends endurance 7%	Nano Energy 2024; TDK 2025 lead-free catalog
Solar Cycle 25 risk	SSN ~115 design basis	SSN ~137 observed; threat assessment +20% elevated	Observed activity exceeded forecast; EMP/GIC exposure during ops elevated	NOAA SWPC Q1 2026
EMI protection Philosophy	Reactive (copper mesh post-event)	Proactive (MXene spray before deployment, dual-function FSS+SE)	Copper mesh corrodes in field; MXene spray permanent and higher SE	ACS Nano 2024 MXene SE survey

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
Frame material	Carbon fiber creates GIC loop; BFRP eliminates it at lower cost per kg
Rotor hub	Al hub couples motor winding GIC to frame; ceramic hub breaks path
EMI shielding on electronics bay	47 dB deficit in V1.0; MXene 2× protection, spray-applied vs mechanical mesh
Navigation	GPS degraded by Solar Energetic Particles during Carrington event
Energy harvesting	Lead-free mandate (RoHS); harvesting extends endurance 7%
Solar Cycle 25 risk	Observed activity exceeded forecast; EMP/GIC exposure during ops elevated
EMI protection Philosophy	Copper mesh corrodes in field; MXene spray permanent and higher SE

Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000102 — Drone Redesign: Archimedes Screw Propulsion V1.0 archived; V2.0 is the active specification as of June 2026.